Biology practice paper

AQA GCSE Combined Science: Trilogy, Biology Paper 1 topics (Foundation tier style)

Time allowed: 30 minutes **Total: 30 marks** **Topics: 4.1 to 4.4**

Name: _____

Instructions

- Answer all the questions in the spaces provided. Use black ink.
- You may use a calculator and a ruler. Show your working in calculations.
- The marks for each question are shown in brackets. Use them to judge how much to write: roughly one mark a minute.
- This is a practice paper written in the style of the real exam. It is not an AQA paper.

Question 1: Cells

1.1 Which structure is found in a plant cell but not in an animal cell? Tick one box. **[1 mark]**

- ☐ Cell membrane
- ☐ Chloroplast
- ☐ Mitochondrion
- ☐ Ribosome

1.2 A student looks at a cheek cell through a light microscope. The image of the cell is 6 mm wide. The real cell is 0.06 mm wide. Calculate the magnification. **[2 marks]**

Magnification = × _____

1.3 Before a cell divides by mitosis, it goes through a stage of growth. Describe two things that happen to the cell during this stage. **[2 marks]**

Question 2: Osmosis practical

A student put cylinders of potato into sugar solutions of different concentrations. She weighed each cylinder before and after leaving it for 30 minutes. The table shows her results.

Concentration of sugar solution in mol/dm³	Percentage change in mass
0.0	+12.0
0.2	+4.5
0.4	-2.0
0.6	-8.5
0.8	-12.5

2.1 One cylinder had a mass of 2.50 g at the start and 2.80 g at the end. Calculate the percentage change in mass. **[2 marks]**

Percentage change = _____ %

2.2 Between which two concentrations in the table would a cylinder show no change in mass? **[1 mark]**

Between _____ and _____ mol/dm³

2.3 Explain why the cylinder in the 0.8 mol/dm³ solution lost mass. **[3 marks]**

2.4 Why did the student calculate the percentage change in mass, rather than just the change in mass? **[1 mark]**

Question 3: Enzymes and food tests

3.1 Name the enzyme that breaks down lipids, and name the products. **[2 marks]**

3.2 Explain why an enzyme stops working at a high temperature. **[2 marks]**

3.3 Name the reagent used to test for protein, and give the colour it turns if protein is present. **[2 marks]**

Question 4: Coronary heart disease

4.1 Explain how a stent helps a person with coronary heart disease. **[2 marks]**

Question 5: Infection and response

5.1 Measles is caused by a virus. Explain why a doctor would not give antibiotics to treat measles. **[1 mark]**

5.2 Explain how a vaccination can stop a person becoming ill with measles. **[3 marks]**

Question 6: Photosynthesis practical

6.1 A student wants to find out how light intensity affects the rate of photosynthesis in pondweed. Describe a method the student could use. Include how they would make it a fair test. **[6 marks]**

End of questions. Check your answers if you have time left.